

Domain Model, Certificate Audit Layer

Certificate (Class)

- **Description:** Represents the current state of an ingested X.509 certificate.
- **Attributes:**
 - id: UUID (Primary Key)
 - serialNumber: String
 - thumbprintSha256: String
 - validTo: Instant
 - auditStatus: String
 - lastAuditedAt: Instant
- **Methods:**
 - Certificate(...) (Constructor for new ingestion)
 - Certificate(...) (Constructor for database loading)
 - setAuditStatus(String)
 - setLastAuditedAt(Instant)
- **Relationships:** Conceptually maps to many AuditLog entries.

AuditLog (Class)

- **Description:** An immutable record representing a single historical audit event.
- **Attributes:**
 - logId: UUID (Primary Key)
 - certificateId: String (Foreign Key)
 - timestamp: Instant
 - statusResult: String
 - details: String (JSONB)
- **Methods:**
 - AuditLog(...) (Constructor for new events)
 - AuditLog(...) (Constructor for database loading)
- **Relationships:** References Certificate by ID.

AuditReport (Record)

- **Description:** An immutable data structure generated by the validation engine.
- **Fields:**
 - isCompliant: boolean
 - violations: List of Strings
 - warnings: List of Strings

2. Data Access Layer (The Repositories)

DatabaseConnectionManager (Class)

- **Description:** Handles raw PostgreSQL connections.
- **Methods:**
 - getConnection()

CertificateRepoInterface (Interface)

- **Methods:**
 - saveCertificate(Certificate)
 - updateAuditState(String id, String status, Instant time)
 - getCertificateById(String)
 - getCertificatesByDomain(String)

CertificateRepository (Class)

- **Relationships:**
 - **Implements:** CertificateRepoInterface
 - **Uses:** DatabaseConnectionManager
 - **Maps to:** Certificate (via internal mapping helper)

AuditLogRepositoryInterface (Interface)

- **Methods:**
 - saveAuditLog(AuditLog)
 - getLogsByCertificateId(String)

AuditLogRepository (Class)

- **Relationships:**
 - **Implements:** AuditLogRepositoryInterface
 - **Uses:** DatabaseConnectionManager
 - **Maps to:** AuditLog (via internal mapping helper)

3. Business Logic & Validation Layer (The Brains)

ComplianceValidator (Interface)

- **Description:** Contract for all compliance checking rules.
- **Methods:**
 - evaluate(Certificate) : returns AuditReport

Nis2BaselineValidator (Class)

- **Relationships:**
 - **Implements:** ComplianceValidator
 - **Produces:** AuditReport

CertificateService (Class)

- **Description:** The main orchestrator for ingesting data.
- **Methods:**
 - ingestCertificate(byte[] rawBytes) : returns Certificate
- **Relationships:**
 - **Depends on:** CertificateRepoInterface
 - **Uses:** CertificateExtractor & CertificateFactory
 - **Uses:** ComplianceValidator (Optional, for Day-Zero checks)

4. Utilities (Stateless Helpers)

CertificateExtractor (Class)

- **Description:** Static utility for mathematical extraction.
- **Methods:**
 - extractThumbprintSha256(X509Certificate)
 - extractKeySize(X509Certificate)

CertificateFactory (Native Java Class)

- **Description:** Built-in Java class for parsing .pem and .cer bytes.
- **Methods:**
 - generateCertificate(InputStream)